Lab 7: Fuzzing for Input Validation Bugs

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Assigned project: SETAPESU26/L6_L7_55_car_insurance_premium. Original source snapshot: 0ef1e60ba3890bc9b75430ab07010cd1a3a151f5.

Results: all three original no-crash properties fail. After validation fixes, all 27 fuzz and regression tests pass. Each of the three required properties uses up to 500 deterministic Hypothesis examples.

Evidence note: screenshots below capture a browser displaying verbatim command output saved from actual executions. They are not native terminal window screenshots. Original text logs are included alongside the images for audit and reproduction.

SS1 and SS3 use the same input: registration ka 01 ab 1234, malformed claims 12000;;800, and DOB 14/08/1999. The buggy program stops at the claims field; the fixed version returns None and continues to the DOB.

SS1 - Original application crash

LAB7 / SS1_buggy_application

```
Captured command output — see README for exact commands. Verbatim log follows.

==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): Traceback (most recent call last):
  File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 39, in <module>
    main()
    ~~~~^^
  File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 32, in main
    print("Total claimed:", total_claim_amount(value))
                            ~~~~~~~~~~~~~~~~~~~^^^^^^^
  File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    return sum(int(amount) for amount in text.split(';'))
  File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 16, in <genexpr>
    return sum(int(amount) for amount in text.split(';'))
               ~~~^^^^^^^^
ValueError: invalid literal for int() with base 10: ''
```

SS2 - Failing Hypothesis tests (part 1)

LAB7 / SS2_failing_fuzz_tests

```
Captured command output — see README for exact commands. Verbatim log follows.

FFF                                                                      [100%]
================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________
test_car_insurance_premium_parser.py:8: in test_normalize_registration_no_crash
    result = normalize_registration(s)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
car_insurance_premium_parser.py:6: in normalize_registration
    plate = plate.strip().upper()
            ^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'strip'

During handling of the above exception, another exception occurred:
test_car_insurance_premium_parser.py:5: in test_normalize_registration_no_crash
    @given(st.text() | st.none())
                   ^^^
test_car_insurance_premium_parser.py:10: in test_normalize_registration_no_crash
    assert False, f"normalize_registration crashed with {s!r}: {e}"
E   AssertionError: normalize_registration crashed with None: 'NoneType' object has no attribute 'strip'
E   assert False
E   Failing test case: test_normalize_registration_no_crash(
E       s=None,
E   )
E   Explanation:
E       These lines were always and only run by failing test cases:
E           /Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py:9
_____________________ test_total_claim_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 14, in
test_total_claim_amount_no_crash
  |     @given(st.text() | st.none())
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 17, in
test_total_claim_amount_no_crash
    |     result = total_claim_amount(s)
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 16, in <genexpr>
    |     return sum(int(amount) for amount in text.split(';'))
    |                ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 19, in
test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with {s!r}: {e}"
    | AssertionError: total_claim_amount crashed with '': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
```

SS2 - Failing Hypothesis tests (part 2)

```
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 17, in
test_total_claim_amount_no_crash
    |     result = total_claim_amount(s)
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |                                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 19, in
test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with {s!r}: {e}"
    | AssertionError: total_claim_amount crashed with None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________ test_driver_age_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 23, in test_driver_age_no_crash
  |     @given(st.text() | st.none())
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 26, in test_driver_age_no_crash
    |     result = driver_age(s)
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 28, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with {s!r}: {e}"
    | AssertionError: driver_age crashed with '': list index out of range
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 26, in test_driver_age_no_crash
    |     result = driver_age(s)
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shreyaskm/Documents/SE/lab-deliverables/LAB7/test_car_insurance_premium_parser.py", line 28, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with {s!r}: {e}"
    | AssertionError: driver_age crashed with None: 'NoneType' object has no attribute 'split'
    | assert False
```

SS2 - Failing Hypothesis tests (part 3)



```
    | Failing test case: test_driver_age_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================= short test summary info ==========================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - Excep...
3 failed in 1.34s
```

SS3 - Fixed application

LAB7 / SS3_fixed_application

```
Captured command output — see README for exact commands. Verbatim log follows.

==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): Total claimed: None

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): Driver age in 2026: 27
```

SS4 - Passing tests



Bug analysis and reflection

The supplied PDF contains no explicit reflection questions, although the repository README requests answers to three. The following three reflections cover the objective, findings and limitations.

1. What bugs did fuzzing uncover?

normalize_registration(None) raises AttributeError because strip is called without a type check. total_claim_amount crashes on None (AttributeError) and empty or malformed tokens (ValueError). driver_age crashes on None (AttributeError) and missing slash-separated fields (IndexError). Hypothesis shrank failing values to None and the empty string. Additional regressions cover impossible dates, future years, negative claims and non-string types.

2. How were they fixed?

Registration checks the input type, returning an empty string for invalid types, then removes whitespace, dashes and dots and uppercases. Claims accepts only nonnegative ASCII integer tokens of at most 100 digits, with optional surrounding whitespace; a malformed record returns None rather than a misleading partial sum. DOB parsing checks DD/MM/YYYY structure, validates the real calendar date and rejects a future birth year. Invalid DOB/current-year inputs return None. Age retains the assigned calendar-year subtraction contract rather than computing birthday-adjusted age.

3. What does fuzzing add, and what are its limits?

Hypothesis explores Unicode, empty strings and None beyond normal examples and shrinks counterexamples for debugging. No-crash tests alone cannot prove correct results: a function returning a constant could pass. Regression examples, a generated sum property, generated valid dates and normalization idempotence check useful semantics too. Passing finite samples is not proof of correctness for all inputs or of resistance to resource exhaustion. Registration normalization does not certify legal plate format. The 100-digit token limit is an explicit validation policy.